

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-2:Lecture-6

Combinational Circuit Design

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Module-2

- ❖ Binary adder and subtractor
- ❖ Ripple carry adders/subtractors and fast adders
- ❖ Binary decoders, encoders
- ❖ Multiplexers and de-multiplexers
- ❖ Logic functions using decoders and multiplexers
- ❖ **Code converters**
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Lecture-6

- ❖ 3-bit Binary to Gray Code
- ❖ 3-bit Gray Code to Binary
- ❖ 4-bit Binary to Gray Code
- ❖ 4-bit Gray Code to Binary

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Code Converter

3-BIT BINARY TO GRAY CODE CONVERSION

Step-1:

Number of Binary inputs = 3

Number of Gray Code output = 3

Step- 2:

The truth table can be obtained as

Binary			Gray Code		
B_2	B_1	B_0	G_2	G_1	G_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

3-BIT BINARY TO GRAY CODE CONVERSION

Step-3: Minimization for the output variables

For G_2

B_1B_0 B_2	00	01	11	10
0				
1	1	1	1	1

B_2

$$G_2 = B_2$$

For G_1

B_1B_0 B_2	00	01	11	10
0			1	1
1	1	1		

B_2B_1' $B_2'B_1$

$$\begin{aligned} G_1 &= B_2B_1' + B_2'B_1 \\ &= B_2 \oplus B_1 \end{aligned}$$

For G_0

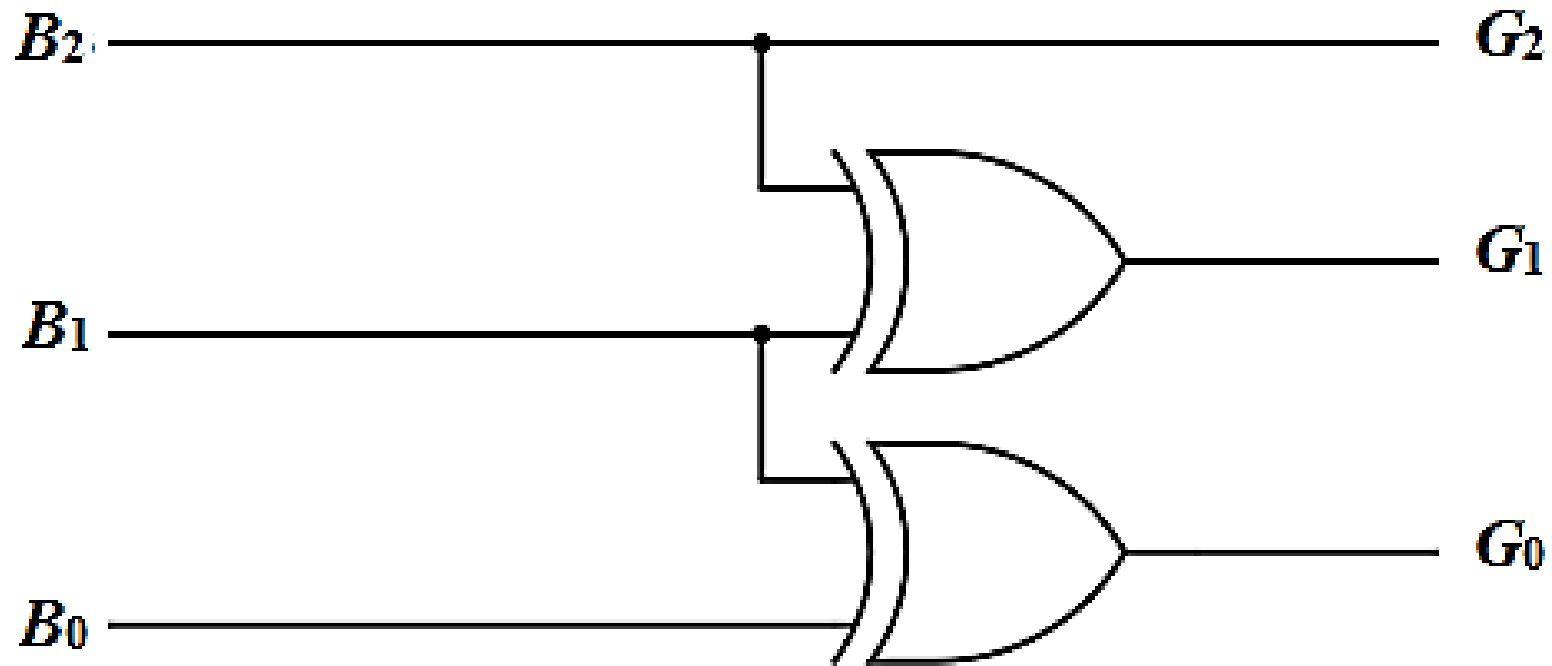
B_1B_0 B_2	00	01	11	10
0		1		1
1		1		1

$B_1'B_0$ B_1B_0'

$$\begin{aligned} G_0 &= B_1B_0' + B_1'B_0 \\ &= B_1 \oplus B_0 \end{aligned}$$

3-BIT BINARY TO GRAY CODE CONVERSION

Step-4: Circuit Design



3-BIT GRAY CODE TO BINARY CONVERSION

Step-1:

Number of Gray Code inputs = 3

Number of Binary output = 3

Step- 2:

The truth table can be obtained as

	Gray Code			Binary		
	G_2	G_1	G_0	B_2	B_1	B_0
0	0	0	0	0	0	0
1	0	0	1	0	0	1
3	0	1	1	0	1	0
2	0	1	0	0	1	1
6	1	1	0	1	0	0
7	1	1	1	1	0	1
5	1	0	1	1	1	0
4	1	0	0	1	1	1

3-BIT GRAY CODE TO BINARY CONVERSION

Step-3: Minimization for the output variables

For B_2

$G_2 \backslash G_1 G_0$	00	01	11	10
0				
1	1	1	1	1

G_2

$$B_2 = G_2$$

For B_1

$G_2 \backslash G_1 G_0$	00	01	11	10
0			1	1
1	1	1		

$G_2 G_1'$ $G_2' G_1$

$$\begin{aligned} B_1 &= G_2 G_1' + G_2' G_1 \\ &= G_2 \oplus G_1 \end{aligned}$$

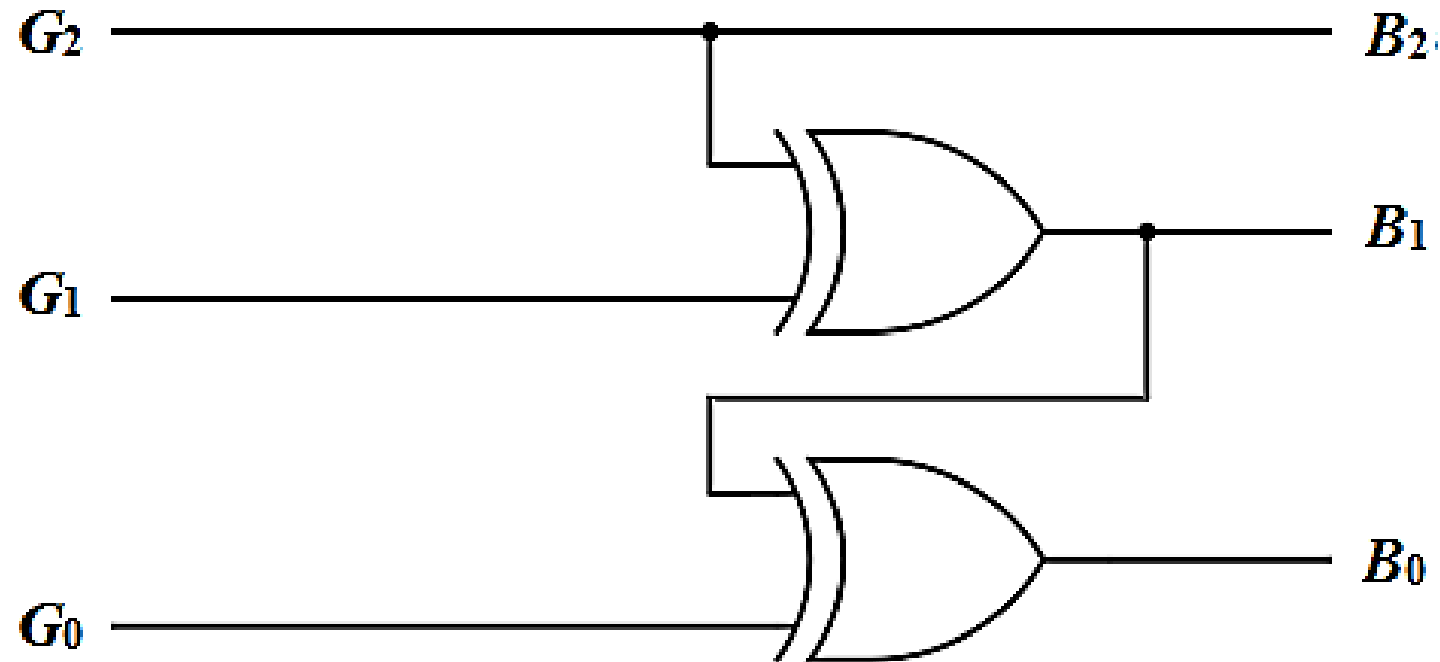
For B_0

$B_2 \backslash B_1 B_0$	00	01	11	10
0		1		1
1	1		1	

$$\begin{aligned} B_0 &= G_2' G_1' G_0 + G_2' G_1 G_0' \\ &\quad + G_2 G_1' G_0' + G_2 G_1 G_0 \\ &= (G_2' G_1 + G_2 G_1') G_0' \\ &\quad + (G_2' G_1' + G_2 G_1) G_0 \\ &= B_1 G_0' + B_1' G_0 = B_1 \oplus G_0 \end{aligned}$$

3-BIT GRAY CODE TO BINARY CONVERSION

Step-4: Circuit Design



4-BIT BINARY TO GRAY CODE CONVERSION

Step-1:

Number of Binary inputs = 4

Number of Gray Code output = 4

Step- 2:

The truth table can be obtained as

Binary				Gray Code			
B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

4-BIT BINARY TO GRAY CODE CONVERSION

Step-3: Minimization for the output variables

For G_3

B_1B_0		B_3B_2			
		00	01	11	10
B_3B_2	00				
	01				
	11	1	1	1	1
	10	1	1	1	1

B_3

$$G_3 = B_3$$

4-BIT BINARY TO GRAY CODE CONVERSION

Step-3: Minimization for the output variables

For G_2

B_1B_0					
B_3B_2		00	01	11	10
00					
01		1	1	1	1
11					
10		1	1	1	1

Diagram illustrating the Karnaugh map for G_2 . The map shows two groups of four 1s:

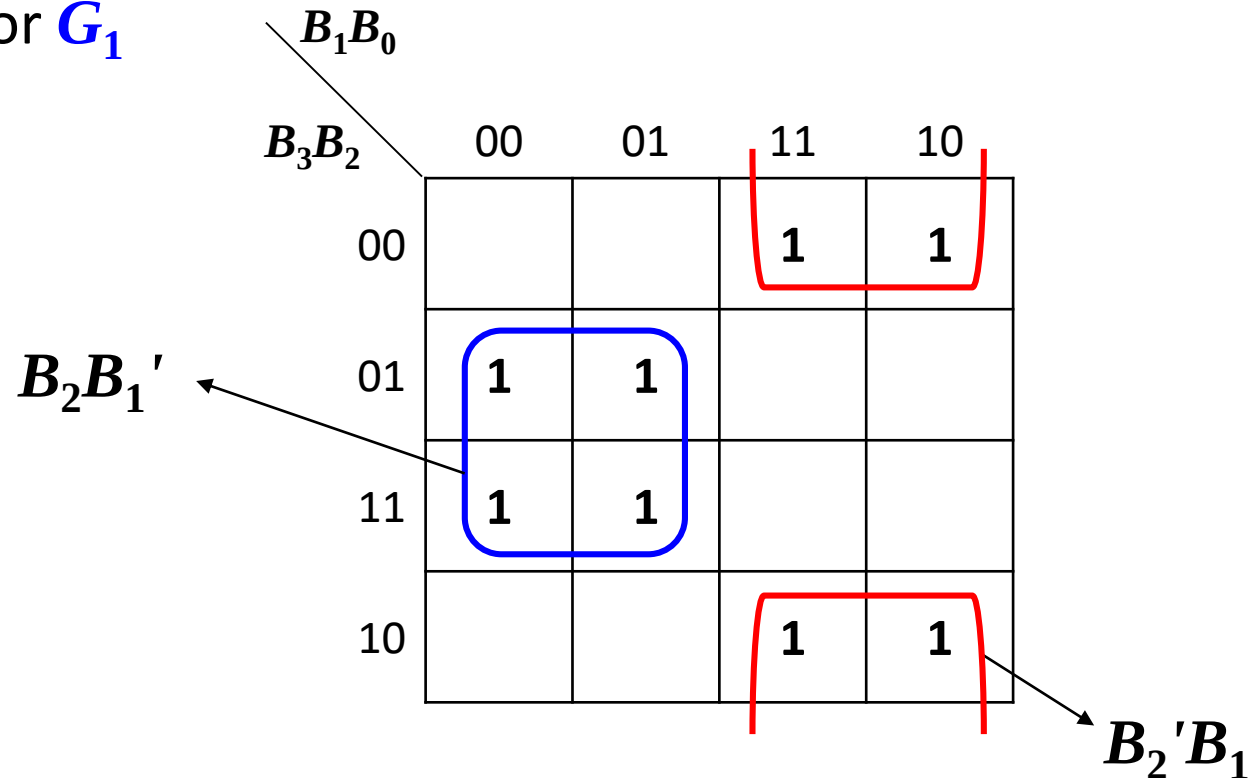
- A blue group labeled $B_3'B_2$ (covering $B_3B_2 = 01$).
- A red group labeled B_3B_2' (covering $B_3B_2 = 10$).

$$\begin{aligned} G_2 &= B_3B_2' + B_3'B_2 \\ &= B_3 \oplus B_2 \end{aligned}$$

4-BIT BINARY TO GRAY CODE CONVERSION

Step-3: Minimization for the output variables

For G_1



$$\begin{aligned} G_1 &= B_2B_1' + B_2'B_1 \\ &= B_2 \oplus B_1 \end{aligned}$$

4-BIT BINARY TO GRAY CODE CONVERSION

Step-3: Minimization for the output variables

For G_0

B_1B_0					
B_3B_2		00	01	11	10
00			1		1
01			1		1
11			1		1
10			1		1

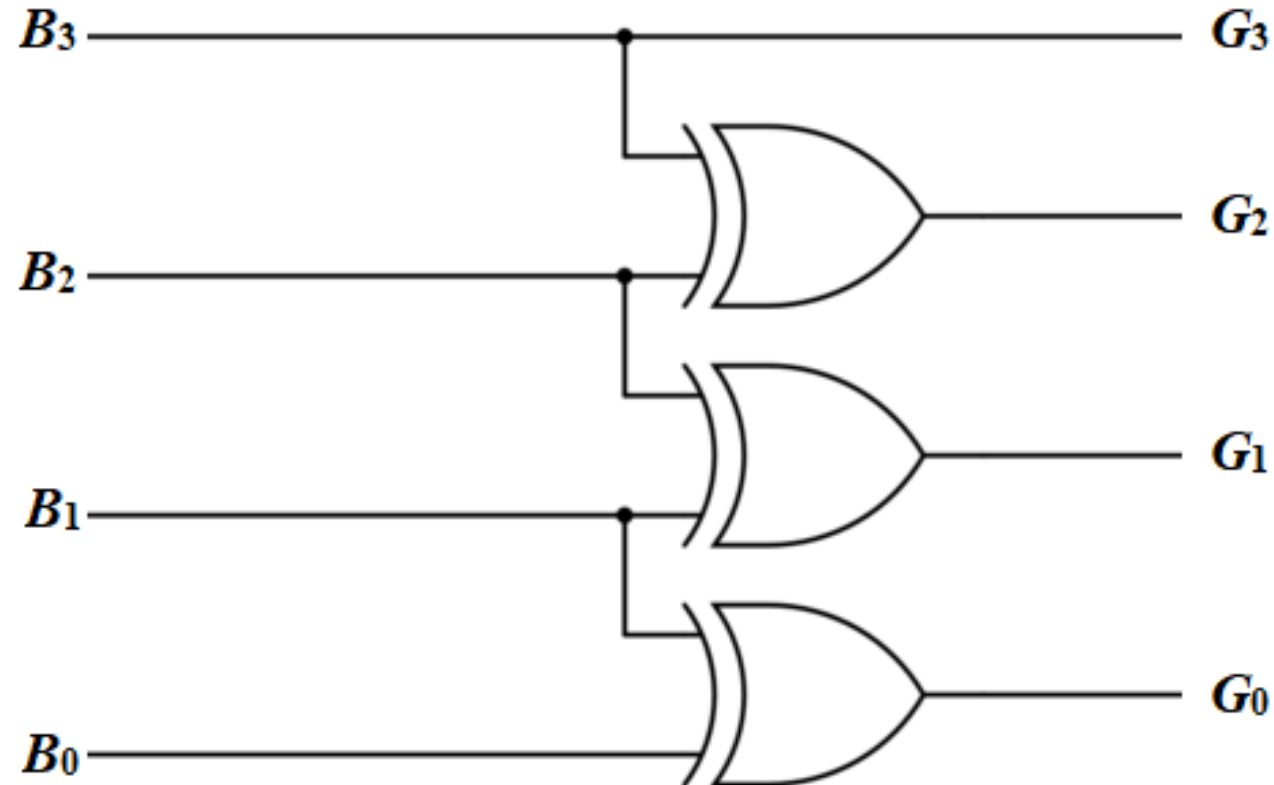
B_1B_0'

$B_1'B_0$

$$\begin{aligned}G_0 &= B_1B_0' + B_1'B_0 \\ &= B_1 \oplus B_0\end{aligned}$$

4-BIT BINARY TO GRAY CODE CONVERSION

Step-4: Circuit Design



4-BIT GRAY CODE TO BINARY CONVERSION

Step-1:

Number of Gray Code inputs = 4

Number of Binary output = 4

Step- 2:

The truth table can be obtained as

	Gray Code				Binary			
	G_3	G_2	G_1	G_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
3	0	0	1	1	0	0	1	0
2	0	0	1	0	0	0	1	1
6	0	1	1	0	0	1	0	0
7	0	1	1	1	0	1	0	1
5	0	1	0	1	0	1	1	0
4	0	1	0	0	0	1	1	1
12	1	1	0	0	1	0	0	0
13	1	1	0	1	1	0	0	1
15	1	1	1	1	1	0	1	0
14	1	1	1	0	1	0	1	1
10	1	0	1	0	1	1	0	0
11	1	0	1	1	1	1	0	1
9	1	0	0	1	1	1	1	0
8	1	0	0	0	1	1	1	1

4-BIT GRAY CODE TO BINARY CONVERSION

Step-3: Minimization for the output variables

For B_3

G_1G_0					
G_3G_2		00	01	11	10
	00				
	01				
	11	1	1	1	1
	10	1	1	1	1

G_3

$$B_3 = G_3$$

4-BIT GRAY CODE TO BINARY CONVERSION

Step-3: Minimization for the output variables

For B_2

G_1G_0					
G_3G_2		00	01	11	10
00					
01		1	1	1	1
11					
10		1	1	1	1

$G_3'G_2$ (points to the blue box)

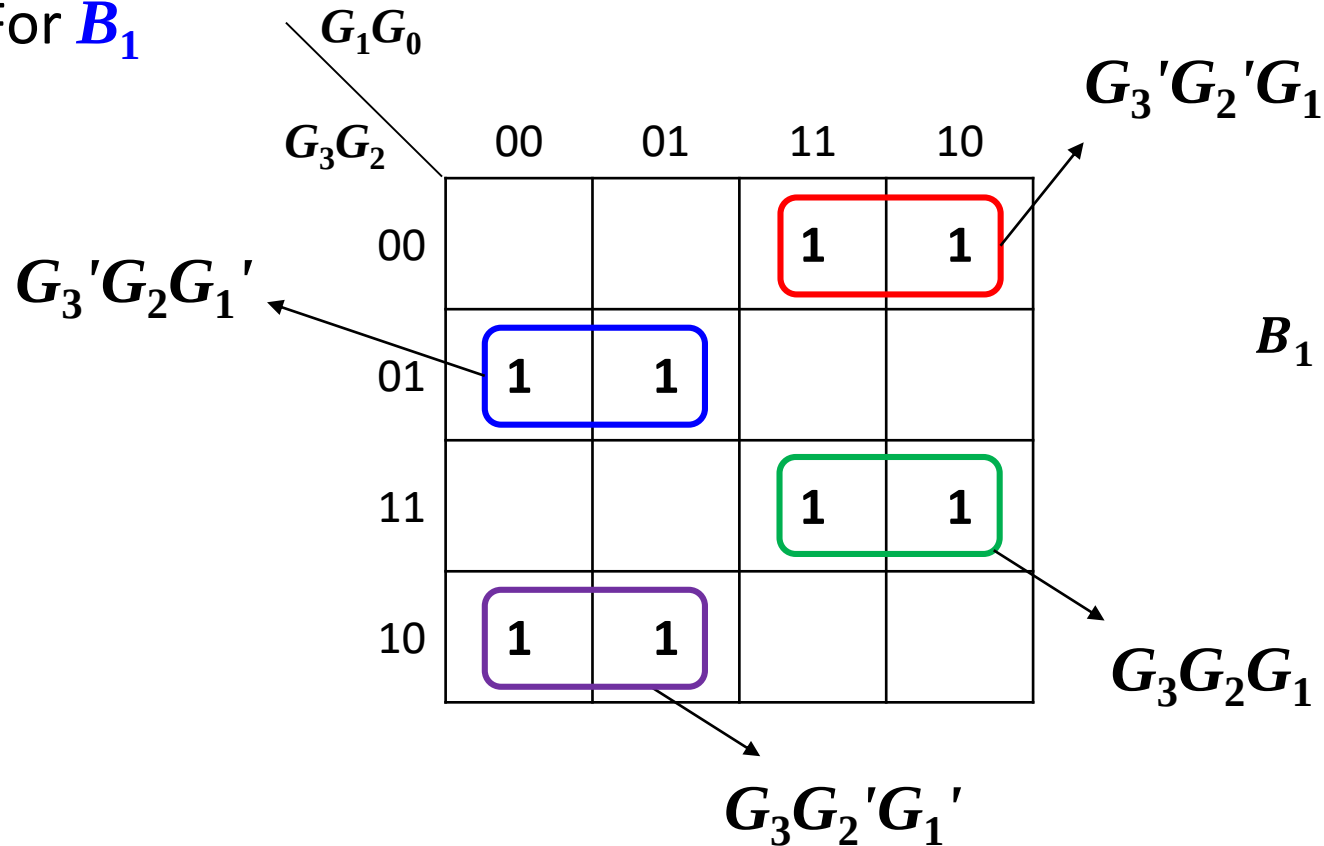
G_3G_2' (points to the red box)

$$\begin{aligned} B_2 &= G_3G_2' + G_3'G_2 \\ &= G_3 \oplus G_2 \end{aligned}$$

4-BIT GRAY CODE TO BINARY CONVERSION

Step-3: Minimization for the output variables

For B_1



$$\begin{aligned}
 B_1 &= G_3'G_2'G_1 + G_3'G_2G_1' + G_3G_2'G_1 + G_3G_2G_1 \\
 &= (G_3'G_2 + G_3G_2')G_1' + (G_3'G_2' + G_3G_2)G_1 \\
 &= B_2G_1' + B_2'G_1 = B_2 \oplus G_1
 \end{aligned}$$

4-BIT GRAY CODE TO BINARY CONVERSION

Step-3: Minimization for the output variables

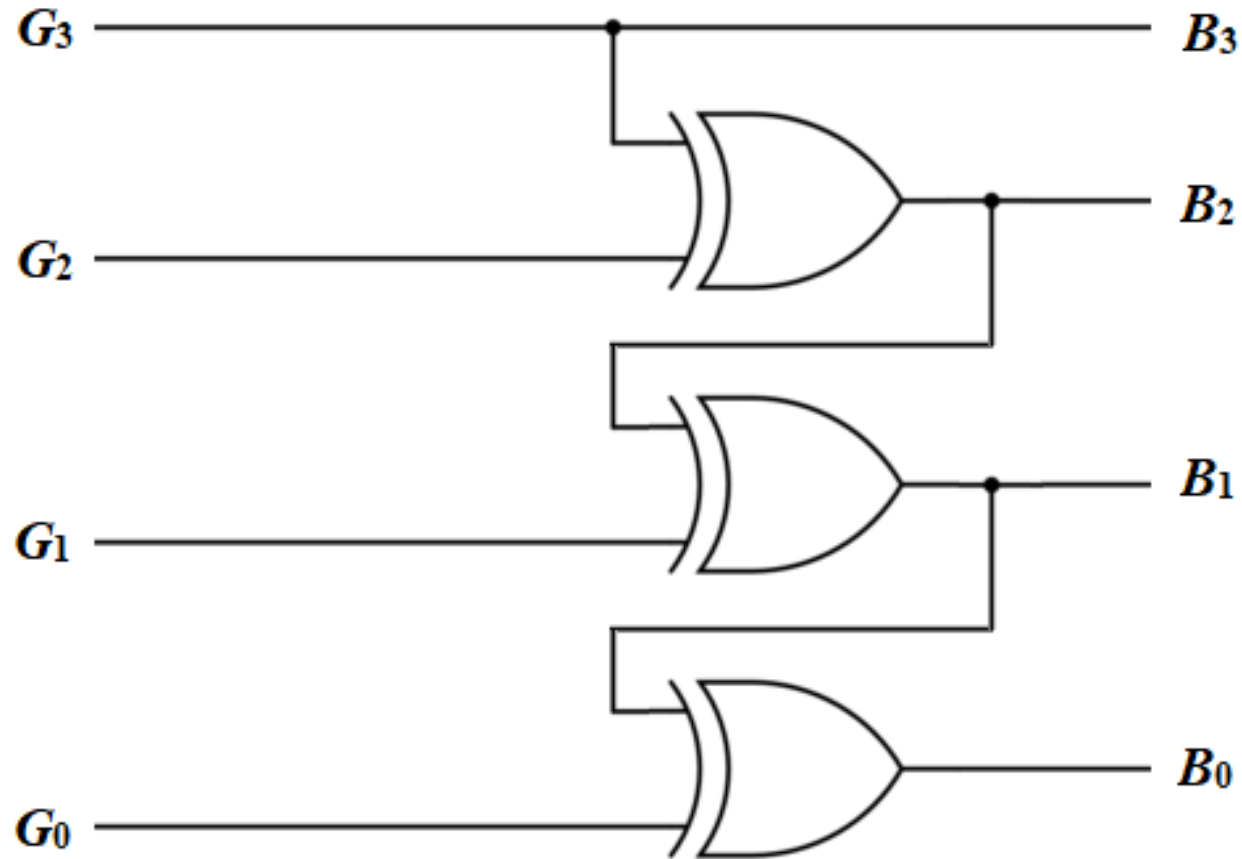
For B_0

G_1G_0				
G_3G_2	00	01	11	10
00		1		1
01	1		1	
11		1		1
10	1		1	

$$\begin{aligned} B_0 &= G'_3G'_2G'_1G_0 + G'_3G'_2G_1G'_0 + G'_3G_2G'_1G'_0 + G'_3G_2G_1G_0 \\ &\quad + G_3G'_2G'_1G'_0 + G_3G'_2G_1G_0 + G_3G_2G'_1G_0 + G_3G_2G_1G'_0 \\ &= (G'_3G'_2G'_1 + G'_3G_2G_1 + G_3G'_2G_1 + G_3G_2G'_1)G_0 \\ &\quad + (G'_3G'_2G_1 + G'_3G_2G'_1 + G_3G'_2G'_1 + G_3G_2G_1)G'_0 \\ &= B'_1G_0 + B_1G'_0 \\ &= B_1 \oplus G_0 \end{aligned}$$

4-BIT GRAY CODE TO BINARY CONVERSION

Step-4: Circuit Design



Thank You